

Credit Name: CSE2920 - CSE Project C

Assignment Name: LED Brightness

How has your program changed from planning to coding to now? Please explain?

PLANNING:

I plan to modify the given code and create a loop that slowly increases and decreases brightness before turning off. I will also Observe how a duty cycle of 0, 0.5, and 1 will affect the LED.

CODING:

1.

```
//Add Phidgets Library
import com.phidget22.*;

public class LEDBrightness {

    public static void main(String[] args) throws Exception{

        //Create
        DigitalOutput redLED = new DigitalOutput();

        //Address
        redLED.setHubPort(1);
        redLED.setIsHubPortDevice(true);

        //Open
        redLED.open(1000);
    }
}
```

Phidgets library is imported

redLED object is created to interact with the physical phidgets box.

The HubPort number was set based on the appropriate object.

IsHubPortDevice set to true for the object (confirming to the program that the objects are part of phidgets box).

Opened the redLED Object, establishing a connection to the physical Phidgets box.

2.

```
//Use your Phidgets with Duty Cycle
while (true) {
    redLED.setDutyCycle(0.0);
    Thread.sleep(150);
    redLED.setDutyCycle(0.2);
    Thread.sleep(150);
    redLED.setDutyCycle(0.4);
    Thread.sleep(150);
    redLED.setDutyCycle(0.6);
    Thread.sleep(150);
    redLED.setDutyCycle(0.8);
    Thread.sleep(150);
    redLED.setDutyCycle(1);
    Thread.sleep(150);
    redLED.setDutyCycle(0.8);
    Thread.sleep(150);
    redLED.setDutyCycle(0.6);
    Thread.sleep(150);
    redLED.setDutyCycle(0.4);
    Thread.sleep(150);
    redLED.setDutyCycle(0.2);
    Thread.sleep(150);
    redLED.setDutyCycle(0);
    Thread.sleep(150);
}
```

While loop is used to repeat the following code.

I slowly increased the duty cycle with increments of 0.2 until I reached 1. Then I did the opposite and went back to a duty cycle of 0.

This creates the effect of a wave, increasing and decreasing brightness over time.

Observations:

Duty cycle number of 0 shows no brightness,

0.5 duty cycle is half brightness

And 1 duty cycle is full brightness.

End Of Program!